

Aliaksei Ivanou | C++ Software Engineer

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Location: Belarus, Minsk | Relocation: Poland (Targeting Work Permit / National Visa D-type paths)

C++ Software Engineer with 6 years of commercial experience in Systems & Embedded development and 7+ years in R&D/DSP. Expert in C++11 - 17, Qt, and Embedded Linux. Proven track record in high-performance system optimization, VoIP, and cross-platform integration (C#, Node.js).

Tech Stack & Domains

- **Primary Languages & Frameworks:** C++ (11-17), C (C89), MATLAB, Qt (5.x/6.x).
- **Interoperability & Scripting:** C# (.NET 8), Node.js, JavaScript.
- **Databases & Storage:** PostgreSQL, MySQL, SQLite.
- **DevOps, Cloud & Infrastructure:** AWS (EC2), Azure (VMs), Jenkins (CI/CD), Yocto (Embedded Linux), Networking: SIP, RTP/SDP, WebRTC, SNMP, TCP/IP, WebSockets.
- **Tools:** Git, CMake, GCC, gdb, Valgrind, Linphone-SDK, FLTK, Google Test, plog.
- **Domains:** VoIP/SIP Telephony, Aerospace & Satellite Systems, UAV Systems, FinTech, Energy Sector ERP, Digital Library Systems.

Experience

C++ Software Engineer | [Innowise Group](#)

12/2024 - Present

Innowise is a global software development company, specializing in Big Data, AI, RPA, IoT, AR/VR, and Blockchain.

- Developed firmware components (C++17 standard used, maintaining legacy Qt 5.15.7 compatibility).
- Extended and customized Linphone-SDK (liblinphone) in C++ to implement SIP failover logic and backup-server connectivity for device reliability.
- Developed signaling logic and UI components for a WebRTC video-conferencing feature; spearheading research into SIP-standard interoperability to support legacy and third-party hardware.
- Configured and deployed mediasoupserver on embedded Linux as a systemd-managed service, including runtime configuration and deployment automation.
- Developed and migrated a .NET-based autoprovisioning utility (v4.8 to .NET 8) to unify project file formats across the ecosystem; implemented device provisioning via SNMP, DHCP, TFTP, and SQLite.
- Collaborated on Yocto/BitBake customization and device-side service configuration for hardware-software integration.
- Technical Lead for a 4-person engineering sub-group: steered code quality, architectural decisions, and onboarding while maintaining 80% individual contributor (IC) output.

Tech Stack & Domains: C++17, Qt 5.15.7 (Widgets, WebSockets), Linphone-SDK, liblinphone, Node.js, mediasoup, libmediasoupclient, WebRTC, SIP/RTP Interoperability, Yocto, BitBake, NXP i.MX, C# (.NET 8), WinForms, SQLite, SNMP, DHCP, TFTP, GitLab, gdb | VoIP/SIP Telephony, Embedded Video Conferencing, Hardware Interoperability.

C++ Software Engineer | [EPAM Systems](#)

05/2020 - 08/2024

(Promoted from Trainee to Mid-level/Senior roles within the 4-year tenure)

EPAM Systems (NYSE: EPAM) is one of the global leaders in both digital engineering and IT consulting, recognized for innovation, cloud, and AI services.

- **FinTech:** Engineered a high-performance Financial Analytics System (C++17) using FLTK and SQLite; established full CI/CD lifecycle with Google Test and CMake.
- Developed C++ modules for two consecutive enterprise projects: an Energy ERP System (C++17) and a legacy Digital Library Platform (C89).
- Managed remote debugging and deployment for C/C++ codebases on AWS EC2 and Azure VMs using SSH and gdb for core dump analysis.
- **Digital Library:** Maintained legacy C89 modules and resolved complex integration defects between the C core and JVM-based services (Scala/Java).
- **Energy ERP:** Optimized PostgreSQL schema and SQL queries, reducing search operation response times from **1.5s to 300ms**.
- Contributed as a Full-stack engineer during delivery phases, maintaining JavaScript modules for data flow between the C++ backend and Web UI.

Tech Stack & Domains: C (C89), C++ (14/17), Java (8/11), Scala, JavaScript, Amazon Web Services (AWS), Microsoft Azure, Apache, Linux, gdb, SSH, git, BitBucket, Jenkins, GCC, Valgrind, CMake, Test-driven development, CI/CD pipelines, PostgreSQL, MySQL, TCP/IP, WebSockets | Energy Sector ERP System, Digital Library Platform.

C++ Software Engineer | [RIFTEK](#)

03/2020 - 05/2020

RIFTEK develops optoelectronic instruments for measuring geometrical quantities, including laser sensors, 2D/3D scanners, and machine vision systems, serving over 70 countries with representation in 45.

- R&D: Developed a 3D reconstruction prototype using Point Cloud Library (PCL) and Qt, facilitating the transition from MATLAB-based research to C++ production code.
- Implemented C++ applications with the Point Cloud Library on Qt.

Tech Stack & Domains: C++ (17), git, GitHub, Qt, Point Cloud Library, Test-driven development, MySQL | Industrial Software, Robotics.

MATLAB Algorithm Developer | [PELENG](#)

08/2012 - 01/2020

PELENG is a developer and manufacturer of high-tech optical and optoelectronic systems for space, industrial, and surveillance applications. The company specializes in thermal imaging, precision optics, and advanced vision systems, emphasizing innovation and collaboration with public and private sectors.

- Developed Earth Remote Sensing algorithms; implemented feature-based image stitching using keypoint matching and defect removal to optimize post-processing workflows.
- Automated resolution assessment for flight tests, replacing manual analysis with MATLAB-based processing modules.
- Implemented a custom lossy image compression algorithm (MATLAB) for satellite transmission, achieving a **reliable 3-4x compression ratio** while carefully balancing integrity preservation and computational constraints.
- Developed software in MATLAB for image type prediction for image quantization, improving image quality metrics by ~30% and enhancing Earth observation accuracy by an estimated 15-20%.
- Developed image processing modules for UAVs (MATLAB), including debayering, stabilization, and sensor fusion.

Tech Stack & Domains: MATLAB, MS Access, git, GitHub, Proprietary Software, TDD, Waterfall | Geoservices, Hardware engineering, Industrial software.

Education

- Bachelor's Degree in Radiophysics. Satellite information systems and technologies: [Belarusian State University, Faculty of Radiophysics and Computer Technologies](#) | 2007 - 2012

Languages

- **English:** Professional Working Proficiency (B2)
- **Polish:** Conversational proficiency (B1)
- **Swedish:** Elementary proficiency (A2)
- **Russian:** Native proficiency (C2)
- **Belarusian:** Native proficiency (C2)